



Configuring the DHCP Service and DHCP Relay Agent to Assign IPv4 Addresses

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

The Dynamic Host Configuration Protocol (DHCP) dynamically configures and uniformly manages IP addresses of hosts. It simplifies network deployment and scale-out, even for small networks.

After completing this task, you will have a clear understanding of the local DHCP address assignment mechanism and be able to configure dynamic and static address assignment using DHCP.

1.2 Objectives

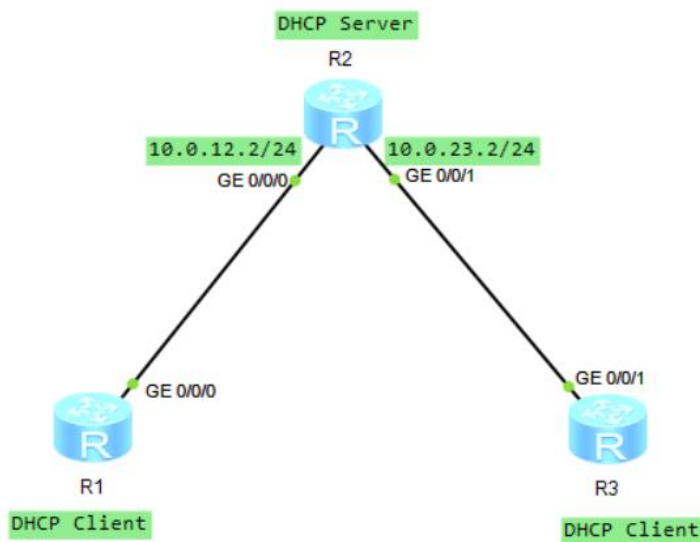
On completing this task, you will be able to:

1. Learn how to configure a DHCP interface address pool.
2. Learn how to configure a DHCP global address pool.
3. Learn how to use DHCP to assign static IP addresses.

2 Task Preparation

2.1 Network Topology

Figure 1 Configuring DHCP to assign IP addresses to hosts



R1 and R3 function as DHCP clients. R2 functions as a DHCP server to assign IP addresses to R1 and R3.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
```

Enter system view, return user view with Ctrl+Z.



```
[Huawei]sysname R1
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.23.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]
```

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 Enabling the DHCP Function

The **dhcp enable** command is the prerequisite for configuring DHCP-related functions including the DHCP client and DHCP server functions. These functions take effect only after the DHCP function is enabled using the **dhcp enable** command.

```
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.

[R2]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.

[R3]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
```



3.3 Configuring an Interface Address Pool

Configure an interface address pool on GigabitEthernet 0/0/0 of R2 to assign an IP address to R1.

```
[R2]interface GigabitEthernet 0/0/0
```

```
[R2-GigabitEthernet0/0/0]dhcp select interface
```

The **dhcp select interface** command enables the DHCP server function on an interface to assign IP addresses from the interface address pool. If you do not run this command, parameters related to the interface address pool cannot be configured.

```
[R2-GigabitEthernet0/0/0]dhcp server dns-list 10.0.12.2
```

The **dhcp server dns-list** command configures the DNS server address for the interface address pool. A maximum of eight DNS server addresses can be configured. These IP addresses are separated by spaces.

3.4 Configuring a Global Address Pool

1. Create an address pool named **GlobalPool**.

```
[R2]ip pool GlobalPool
```

Info: It's successful to create an IP address pool.

```
[R2-ip-pool-GlobalPool]network 10.0.23.0 mask 24
```

The **network** command configures a network segment where addresses can be assigned for the global address pool.

```
[R2-ip-pool-GlobalPool]dns-list 10.0.23.2
```

```
[R2-ip-pool-GlobalPool]gateway-list 10.0.23.2
```

The **gateway-list** command configures the egress gateway address for DHCP clients. After obtaining an address, R3 generates a default route with the next-hop address 10.0.23.2.

```
[R2-ip-pool-GlobalPool]lease day 2 hour 2
```

The **lease** command configures the lease for IP addresses in the address pool. If the lease is set to **unlimited**, the lease is unlimited. By default, the IP address lease is one day.

```
[R2-ip-pool-GlobalPool]static-bind ip-address 10.0.23.3 mac-address 00e0-fc88-4b5f
```

The **static-bind** command binds an IP address in the global address pool of the DHCP server to a MAC address. 00e0-fc88-4b5f is the MAC address of GigabitEthernet0/0/1 on R3 in the current lab environment. You can run the **display interface GigabitEthernet0/0/1** command on R3 to view the MAC address of the interface. After the command is configured, R3 obtains the fixed IP address 10.0.23.3.

```
[R2-ip-pool-GlobalPool]quit
```

3.5 Assigning a Static IP Address to R3

Enable the DHCP server function on GigabitEthernet 0/0/1 of R2 to assign an IP address to R3.

```
[R2]interface GigabitEthernet 0/0/1
```

```
[R2-GigabitEthernet0/0/1]dhcp select global
```



The **dhcp select global** command enables the DHCP server function on an interface to assign IP addresses from the global address pool. After receiving a request from a DHCP client, the interface searches the global address pool for an available IP address and assigns the IP address to the DHCP client.

3.6 Configuring DHCP Clients

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0] ip address dhcp-alloc
[R1-GigabitEthernet0/0/0]quit
[R1]

[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1] ip address dhcp-alloc
[R3-GigabitEthernet0/0/1]quit
[R1]
```

4 Verification

4.1 Checking Addresses and Routes of R1 and R3

- Check information on R1.

```
[R1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.12.254/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	down	down
NULL0	unassigned	up	up(s)

```
[R1]
```

```
[R1]display dns server
Type:
```



D:Dynamic S:Static

No. Type IP Address
1 D 2.12.0.10

No configured ipv6 dns servers.

[R1]

[R1]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 8 Routes : 8

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Unr	60	0	D	10.0.12.2	GigabitEthernet 0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.254	GigabitEthernet 0/0/0
10.0.12.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/0
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R1]

- Check information on R3.

[R3]display ip int brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 2

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 2

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down



```
GigabitEthernet0/0/1          10.0.23.3/24          up          up
GigabitEthernet0/0/2          unassigned            down        down
NULL0                         unassigned            up          up(s)
[R3]

[R3]display dns server
Type:
D:Dynamic      S:Static

No.  Type  IP Address
1    D     2.23.0.10

No configured ipv6 dns servers.
[R3]

[R3]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
          Destinations : 8          Routes : 8

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
          0.0.0.0/0    Unr      60    0              D    10.0.23.2    GigabitEthernet 0/0/1
          10.0.23.0/24 Direct    0     0              D    10.0.23.3    GigabitEthernet 0/0/1
          10.0.23.3/32 Direct    0     0              D    127.0.0.1    GigabitEthernet 0/0/1
          10.0.23.255/32 Direct    0     0              D    127.0.0.1    GigabitEthernet 0/0/1
          127.0.0.0/8  Direct    0     0              D    127.0.0.1    InLoopBack0
          127.0.0.1/32 Direct    0     0              D    127.0.0.1    InLoopBack0
          127.255.255.255/32 Direct    0     0              D    127.0.0.1    InLoopBack0
          255.255.255.255/32 Direct    0     0              D    127.0.0.1    InLoopBack0
[R3]
```

4.2 Checking Address Assignment on R2

```
<R2>display ip pool name GlobalPool

Pool-name       : GlobalPool
Pool-No         : 1
Lease           : 2 Days 2 Hours 0 Minutes
Domain-name     : -
```



```
DNS-server0      : 10.0.23.2
NBNS-server0     : -
Netbios-type     : -
Position         : Local           Status           : Unlocked
Gateway-0        : 10.0.23.2
Mask             : 255.255.255.0
VPN instance     : --
```

Start	End	Total	Used	Idle(Expired)	Conflict	Disable
10.0.23.1	10.0.23.254	253	1	252(0)	0	0

<R2>

The **display ip pool** command displays information about a configured IP address pool, including the address pool name, lease, lock status, and IP address status.

<R2>display ip pool interface GigabitEthernet0/0/0

```
Pool-name        : GigabitEthernet0/0/0
Pool-No          : 0
Lease            : 1 Days 0 Hours 0 Minutes
Domain-name      : -
DNS-server0      : 10.0.12.2
NBNS-server0     : -
Netbios-type     : -
Position         : Interface       Status           : Unlocked
Gateway-0        : 10.0.12.2
Mask             : 255.255.255.0
VPN instance     : --
```

Start	End	Total	Used	Idle(Expired)	Conflict	Disable
10.0.12.1	10.0.12.254	253	1	252(0)	0	0



<R2>

When an interface address pool is configured, the name of the address pool is the name of the interface. The assigned gateway address is the IP address of the interface and cannot be changed.

5 Quiz

1. What are the differences between the application scenario of a global address pool and that of an interface address pool?
2. If there are multiple global address pools, how do I determine the global address pool from which an IP address is assigned to a DHCP client?
3. Which IP addresses in an address pool are reserved?

In an IP address pool, the IP addresses assigned to servers such as the DNS server and the IP address of the DHCP server interface should be excluded to avoid IP address conflicts.